

Summer 2026 GRAD 505 – Foundations in Data Science

Week 6

Lab 4 Risk Assessment in Food Science

Authored by:

Bradley May

Question 1 - Which group is most at risk? With the risk values you have calculated for each test subject you can now answer some questions about that risk. Calculate the mean and standard deviation of the risk. There is much variation in risk.

Across all 7,267 subjects, the average relative exposure is 16.81% of the Disulfoton reference dose, with a standard deviation of 26.86%. The fact that the standard deviation is larger than the mean tells us immediately that there is a wide spread in risk across the population, some people have very low exposure while others have quite high exposure. This is confirmed by the range, which runs from 0% to 798%.

Looking more closely at the distribution, the median is 7.91%, which is less than half the average. This means most subjects have relatively low exposure, but a smaller group of high-consumption individuals pulls the average upward. In practical terms, the average alone does not tell the full story here, the typical subject's exposure is better represented by the median. About 23% of subjects consumed neither vegetable and have zero exposure, which contributes to the wide spread. On the other end, roughly 1 in 60 subjects (1.67%) exceeded 100% of the reference dose individually.

The Female under-10 group is most at risk, with an average relative exposure of 20.48% (SD = 31.93%, n = 878). The reason children rank highest comes down to body weight. The exposure calculation divides the amount of pesticide consumed by the subject's body weight, so a child eating the same amount of vegetables as an adult receives a higher dose relative to their size. Adults 40 and older rank lowest (around 12% for both sexes), which reflects their larger body mass spreading the same dose over more body weight.

No group exceeds the reference dose on average, which is a reassuring finding. That said, a few limitations are worth noting. The analysis uses average pesticide concentrations applied to every subject, which does not capture variability between individual vegetables purchased at different markets or times of year. The analysis also only covers two vegetables and five pesticides, so it likely underestimates total dietary pesticide exposure. The individuals most worth paying attention to are the small percentage of young, lower-weight, high-consumption subjects who do exceed the reference dose individually.

Question 2 - Could sex and age class of the test subjects be part of the reason for this variation, i.e., if all subject were the same sex and same age, would the variation be greater or lesser than is present with this current data?

Sex and age class do contribute to the variation in risk, but only to a very small degree. When we separate the data into the eight groups, the differences in group averages account for only about 1.5% of the total variation in the dataset. The remaining 98.5% of the variation exists within the groups themselves meaning people of the same sex and age still vary enormously in their exposure.

So to directly answer the question: if all subjects were the same sex and age, the variation would be slightly lesser than what we see in the full dataset, but only marginally so. The overall standard deviation is 26.86%, and the average standard deviation within any single group is 26.68%, nearly identical. Isolating one group would not meaningfully reduce the spread.

What this tells us is that sex and age are not the primary drivers of variation in this dataset. The bigger factor is individual dietary behavior, specifically, how much of these two vegetables a person actually eats and how often. A child who rarely eats cabbage or Chinese kale will have far lower exposure than another child of the same age and sex who eats them frequently, and that difference swamps any pattern explained by demographics alone.

One interesting exception worth noting: the Male under-10 group has a within-group standard deviation of 40.26%, well above the overall average. This suggests that consumption habits among young boys are especially inconsistent, some eat these vegetables very frequently while others rarely do, making this group particularly difficult to characterize with a single average risk value.

Question 3 - Calculate the mean risk within each of the eight groups. Which group is most at risk?

The table below shows mean relative exposure (% of reference dose) for all eight groups, ranked highest to lowest.

Rank	Group	Mean % of Ref Dose	SD (%)	n
1	Female <10	20.48%	31.93%	878
2	Male <10	19.07%	40.26%	949
3	Female 18-39	18.95%	22.74%	715
4	Female 10-17	18.94%	26.46%	931
5	Male 18-39	18.90%	23.93%	618
6	Male 10-17	17.31%	22.21%	916
7	Female 40+	12.13%	21.97%	1201
8	Male 40+	12.08%	17.68%	1059

The Female under-10 group carries the highest mean risk at 20.48% of the reference dose. As discussed in Question 1, this comes down to body weight, young children receive a higher dose per kilogram of body weight compared to older subjects consuming the same amount of food. It is also worth noting that the top six groups are close to one another in their averages, ranging from about 17% to 20%, while adults 40 and older drop noticeably to around 12%. This suggests age has a clearer influence on risk than sex does, since the male and female rankings are mixed throughout the table rather than one sex consistently outranking the other.